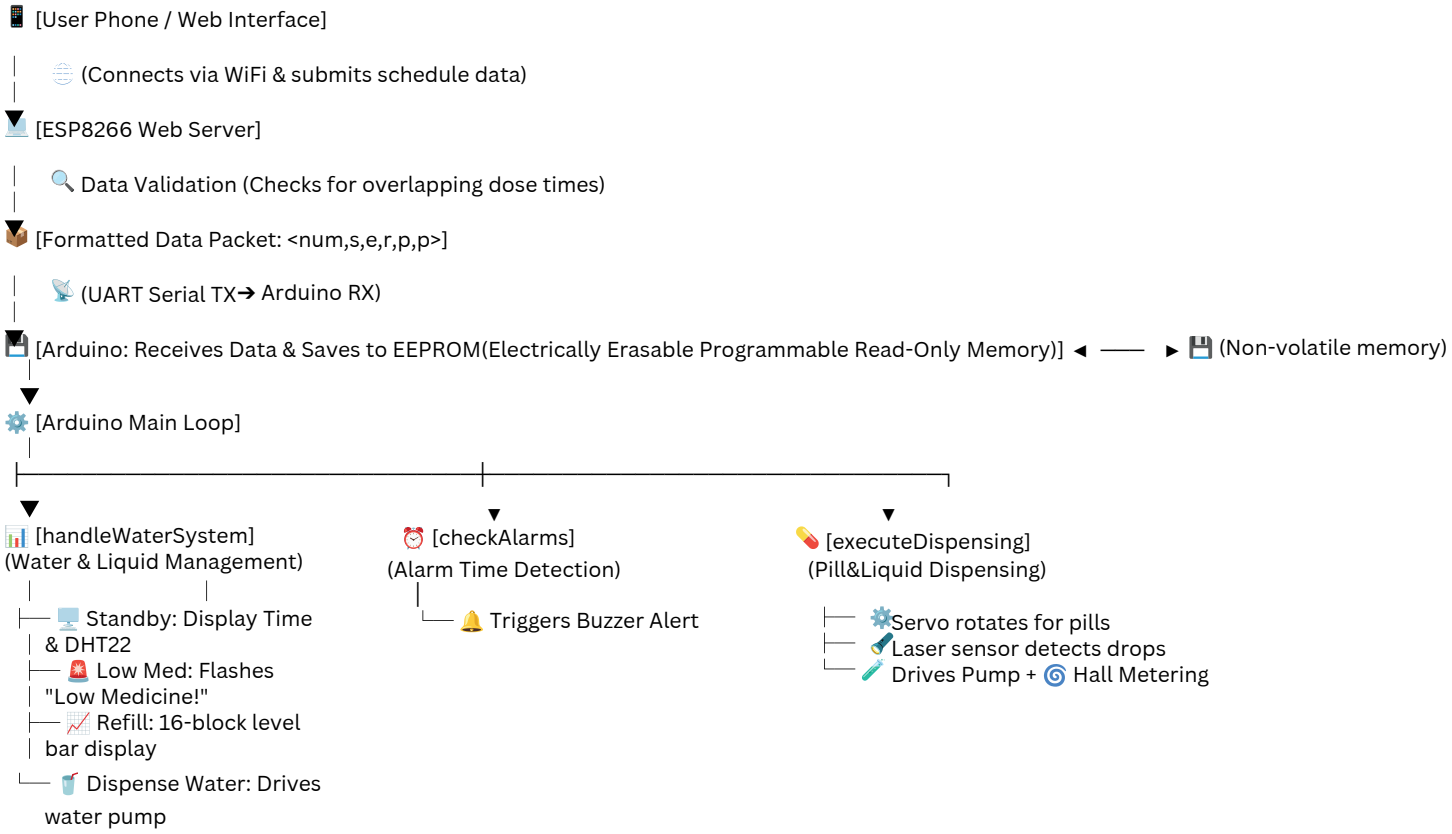
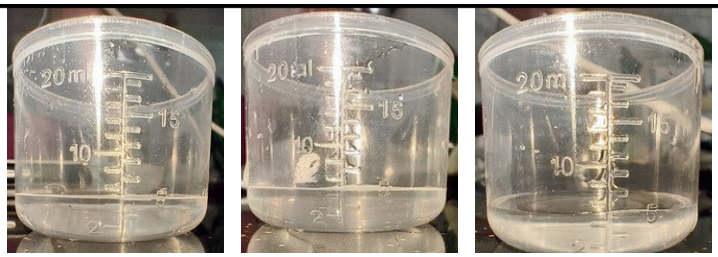
Technical Documentation & Calibration Logbook

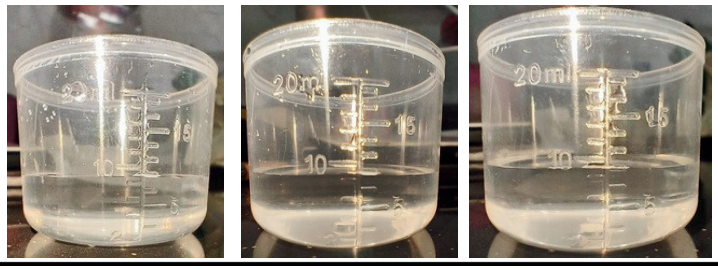

[User Phone / Web Interface]
(Connects via WiFi & submits schedule data)
[ESP8266 Web Server]
Data Validation (Checks for overlapping dose times)
[Formatted Data Packet: <num,s,e,r,p,p>]
(UART Serial TX➔ Arduino RX)
[Arduino: Receives Data & Saves to EEPROM(Electrically Erasable Programmable Read-Only Memory)]
(Non-volatile memory)
[Arduino Main Loop]
[handleWaterSystem]
(Water & Liquid Management)
Standby: Display Time & DHT22
Low Med: Flashes "Low Medicine!"
Refill: 16-block level bar display
Dispense Water: Drives water pump
[checkAlarms]
(Alarm Time Detection)
Triggers Buzzer Alert
[executeDispensing]
(Pill&Liquid Dispensing)
Servo rotates for pills
Laser sensor detects drops
Drives Pump + Hall Metering

Component / Module	Pin	Arduino Pin / Connection	Description / Function
ESP8266 WiFi	TX	Arduino D2 (PIN_ESP_RX)	Software Serial RX (Receives data from ESP8266)
	RX	Arduino D3 (PIN_ESP_TX)	Software Serial TX (Sends data to 3.3V RX via voltage
	VCC / GND	3.3V / GND	Power supply and common ground
LCD 1602 (I2C)	SDA	Arduino A4	I2C Data Line
	SCL	Arduino A5	I2C Clock Line
	VCC / GND	5V / GND	Power supply and common ground
Medicine Level Sensor	Signal	Arduino A0 (PIN_MED_LEVEL)	Analog medicine tank level measurement
DHT Temp & Humidity Sensor	DATA	Arduino A1 (PIN_DHT)	Single-bus temperature & humidity data input
Water Level Sensor	Signal	Arduino A3 (PIN_WATER_LEVEL)	Analog drinking water tank level measurement
Main Dispense Button	Signal	Arduino D4 (PIN_MAIN_BUTTON)	Manual pill dispensing trigger input
Medicine Pump Relay	IN1	Arduino D5 (PIN_MED_PUMP)	Control signal for liquid medicine pump
Water Fill Limit Switch	Signal	Arduino D6 (PIN_FILL_SWITCH)	Drinking water tank refill detection input
Medicine Fill Limit Switch	Signal	Arduino D7 (PIN_MED_FILL_SWITCH)	Medicine tank refill detection input
Hall Flow Sensor	Signal	Arduino D8 (PIN_HALL)	Pulse counting for liquid medicine dispensing
Water Dispense Button	Signal	Arduino D9 (PIN_WATER_BUTTON)	Manual drinking water dispense trigger input
Laser Sensor	Signal	Arduino D10 (PIN_LASER)	Optical beam drop detector for pill passage
Servo Motor	PWM	Arduino D11 (PIN_SERVO)	PWM signal for pill dispensing mechanism
Water Pump Relay	IN2	Arduino D12 (PIN_WATER_PUMP)	Control signal for drinking water pump
Buzzer	Signal	Arduino D13 (PIN_BUZZER)	Audio alert and alarm output

data collected of hall sensor realiblity ,accuracy and logbook for ratio of pulse and volume (to find the relationship between hall sensor pulse and solution volume)

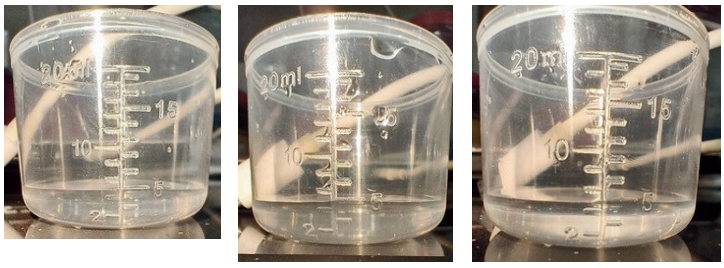


650 pulse

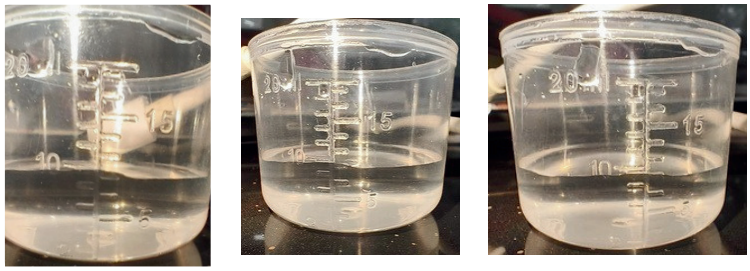


1300 pulse

conclude:650<5 ml

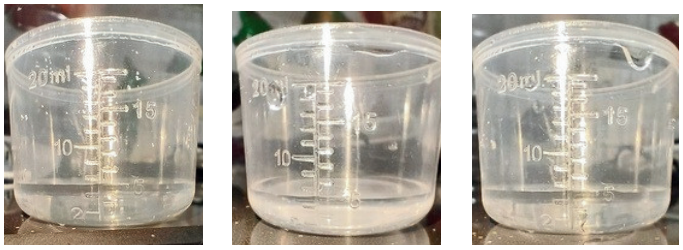


700 pulse

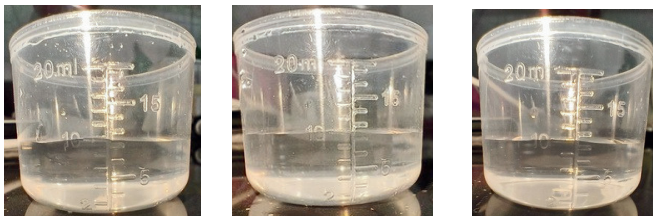


1400 pulse

conclude:700<5 ml

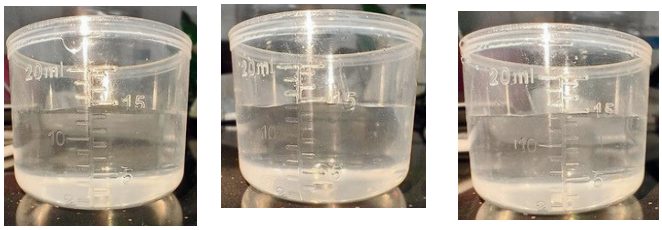


750 pulse

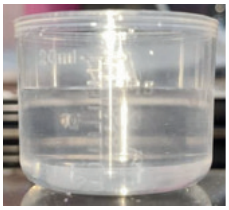
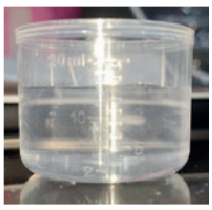


1500 pulse

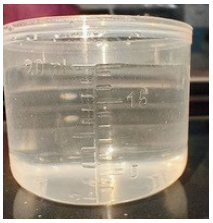
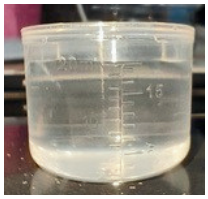
conclude:750<5 ml



2150 pulse

2400pulse

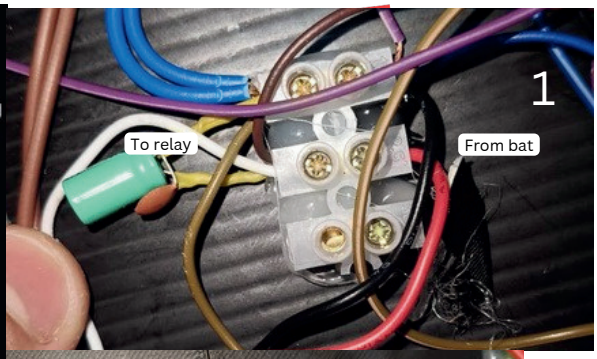
3200pulse

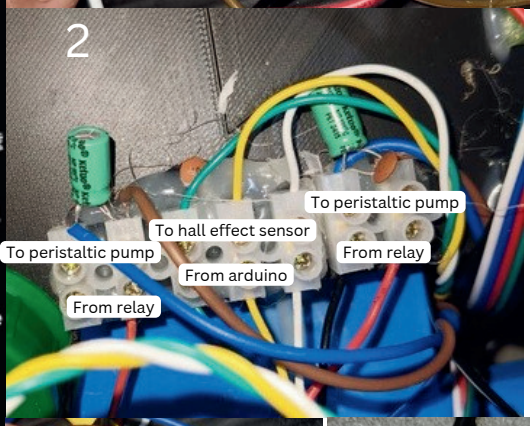
conclude:3200≈20ml
2400≈15ml

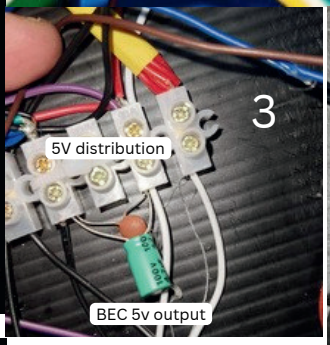
in conclusion 3200/20=160
from the experiment we comfirm that 160 hall pulse is approximately equals to 1 ml

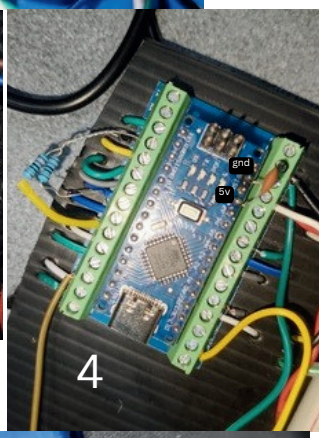
Power & Capacitive Filtering Architecture（电源与电容滤波架构）

- **Inductive Load Suppression (Pump Output) / 感性负载抑制（水泵输出端）**
  - **EN:** Water pumps operate as inductive loads, drawing massive inrush currents during startup and generating high-voltage spikes via back-electromotive force (Back-EMF) upon shutdown. Parallel capacitors buffer sudden current surges and clamp voltage spikes to prevent MCU brownouts.
  - **ZH:** 水泵属于感性负载，启动瞬间会产生极大的冲击电流（Inrush Current），关断时会通过反向电动势（Back-EMF）产生高压尖峰。并联电容能缓冲突增电流并钳制高压尖峰，防止单片机因电压骤降而欠压复位（Brownout）。
- **Switching Noise Attenuation (BEC Output) / 开关噪声衰减（BEC 输出端）**
  - **EN:** Battery Elimination Circuits (BEC) utilize high-frequency switching (hundreds of kHz), introducing voltage ripple. Output capacitors filter out this high-frequency switching noise to supply a stable 5V DC rail for microcontrollers.
  - **ZH:** BEC 稳压模块采用数百 kHz 的高频开关工作，会引入电压纹波。输出端电容能滤除这种高频开关噪声，为单片机提供平稳的 5V 直流供电轨。
- **Dual-Capacitor Topology (Electrolytic + Ceramic) / 双电容组合拓扑（电解电容 + 陶瓷电容）**
  - **EN:** Combines bulk storage and high-frequency filtering. High-capacity electrolytic capacitors ($100\ \mu F$) absorb low-frequency voltage sags and current surges (despite higher ESR/ESL). Parallel ceramic capacitors ($0.1\ \mu F$) feature near-zero ESR/ESL, instantly shunting high-frequency voltage transients.
  - **ZH:** 将大容量储能与高频滤波结合。大容量电解电容（$100\ \mu F$）虽然 ESR/ESL 较高，但能吸收低频电压骤降和大电流冲击；并联的陶瓷电容（$0.1\ \mu F$）具有近乎为零的 ESR/ESL，能瞬间将高频电压瞬态噪声旁路泄放。
- **MCU Local Decoupling (Arduino 5V/GND) / 单片机本地去耦（Arduino 5V/GND）**
  - **EN:** Since bulk power filtering is handled upstream at the BEC, the Arduino requires only a local $0.1\ \mu F$ ceramic decoupling/bypass capacitor placed directly across its power pins to suppress internal digital logic switching noise.
  - **ZH:** 由于主电源低频滤波已在前端 BEC 完成，Arduino 仅需在电源引脚旁就近跨接一颗 $0.1\ \mu F$ 陶瓷去耦/旁路电容，即可有效抑制其内部数字逻辑开关产生的噪声。



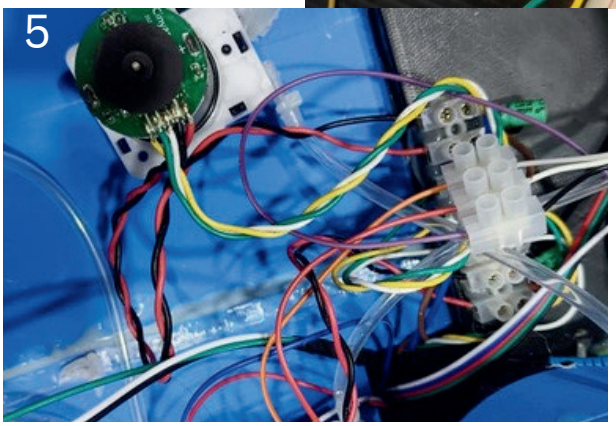


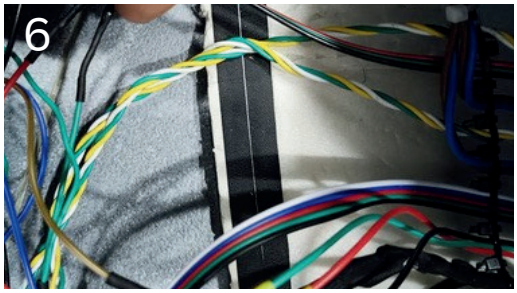




Wiring & EMI Suppression Strategy（线缆布线与电磁干扰抑制策略）

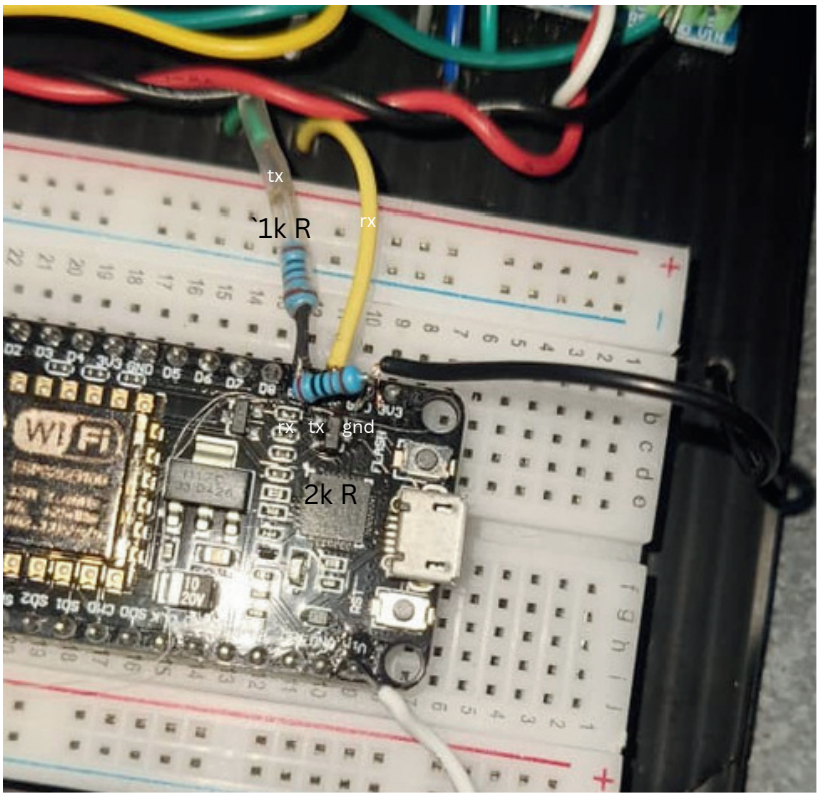
- **Motor Twisted-Pair Wiring / 电机线双绞布线**
  - **EN:** As high-power inductive loads with rapid current transitions ($\frac{di}{dt}$), motor lines emit severe Electromagnetic Interference (EMI). Twisting the positive and negative wires forces equal and opposite currents, effectively canceling radiated magnetic fields.
  - **ZH:** 电机作为大功率感性负载，具备极高的电流变化率（$\frac{di}{dt}$），其导线会发射严重的电磁干扰（EMI）。将正负极线绞合可强制使电流大小相等、方向相反，从而有效抵消向外辐射的磁场。
- **Hall Sensor 3-Wire Twist (Signal + VCC + GND) / 霍尔传感器三线双绞（信号线 + VCC + GND）**
  - **EN:** Hall sensors transmit high-frequency pulse trains vulnerable to motor noise and false count interrupts. Twisting Signal, VCC, and GND minimizes the signal loop area and forces inductive noise to act as common-mode noise, protecting pulse integrity.
  - **ZH:** 霍尔传感器传输的高频脉冲信号极易受到电机噪声干扰而产生误计数中断。将信号线、VCC 与 GND 绞合可最小化信号回路面积，并使感应噪声转化为共模噪声，确保脉冲计数的完整性。
- **Motor 2-Wire Twist (VCC + GND) / 电机双线双绞（VCC + GND）**
  - **EN:** Motors act purely as power-delivery actuators without signal lines, requiring only a 2-wire twist to cancel differential-mode radiated emissions.
  - **ZH:** 电机纯粹作为功率传输执行机构而无信号线，仅需正负电源双线绞合即可抵消差模辐射发射。
- **Standard Wiring for Switches & LCDs / 按键开关与 LCD 的标准布线**
  - **EN:** Buttons and switches handle static DC signals stabilized via software debouncing. LCD lines operate with short trace lengths and strong noise margins. Neither acts as a high-frequency pulse source nor draws inductive current, making twisted-pair wiring redundant.
  - **ZH:** 按键与开关传输由软件防抖稳定后的静态直流电平；LCD 走线短且具备较强的噪声容限。两者既不是高频脉冲源也不抽取感性电流，因此无需使用双绞线。






```
if (digitalRead(PIN_MAIN_BUTTON) == HIGH) {
    delay(50); // 👆 软件滤波的核心：避开微秒级的电电平毛刺
    if (digitalRead(PIN_MAIN_BUTTON) == HIGH) {
        // 确认高电平持续了 50ms 以上，才认定是人为真正按下了按键
    }
}
```

7



These two resistors form a **Voltage Divider** circuit to step down the 5V signal from the Arduino to 3.3V, preventing damage to the ESP8266 RX pin.

Core Reason

- The TX output of the Arduino (such as Uno / Nano) is 5V when outputting a HIGH logic level.
- The operating voltage and IO pin tolerance limit of the ESP8266 chip is 3.3V (it cannot directly withstand 5V).

Circuit Operating Principle and Calculation

Using $1\text{k}\Omega$ and $2\text{k}\Omega$ resistors for voltage division, the formula is as follows:

$$V_{out} = V_{in} \times \frac{R_2}{R_1 + R_2}$$

- $V_{in} = 5\text{V}$ (from Arduino TX)
- $R_1 = 1\text{k}\Omega$ (connected in series between Arduino TX and ESP8266 RX)
- $R_2 = 2\text{k}\Omega$ (connected between ESP8266 RX and GND)

Calculation:

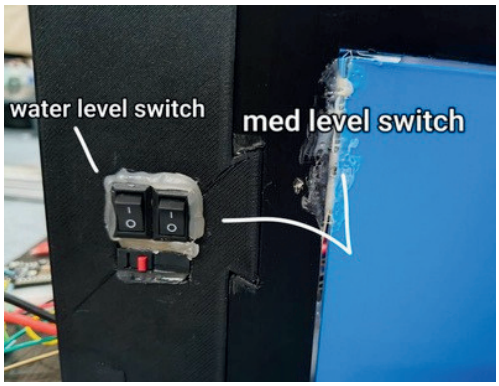
$$V_{out} = 5\text{V} \times \frac{2\text{k}\Omega}{1\text{k}\Omega + 2\text{k}\Omega} = 5\text{V} \times \frac{2}{3} \approx 3.33\text{V}$$

The stepped-down 3.33V perfectly fits the standard HIGH logic level range of the ESP8266 RX pin.

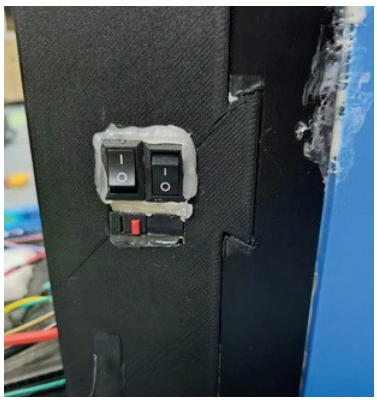
Additional Communication Details

- **Arduino TX $\rightarrow$ ESP8266 RX:** Must include a voltage divider circuit (steps down 5V to 3.3V).
- **ESP8266 TX $\rightarrow$ Arduino RX:** No resistors are required; a direct connection can be used because the 3.3V output from the ESP8266 already exceeds the minimum threshold for Arduino to register a HIGH level (approx. 3.0V).

operator guide



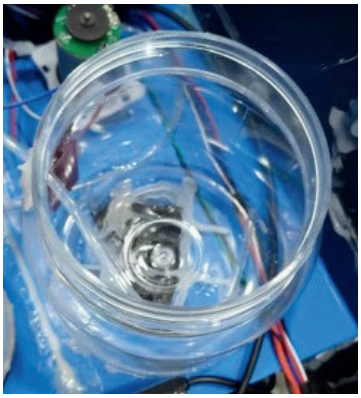
Safety Interlock: Pump activity automatically disables if liquid level falls below critical threshold (<50) to prevent dry burning.



Flip the water level switch to on position



Ensure the water Level progress bar indicates normal operational status.



This is the drinking water tank fill in with water



LCD progress bar indicates real-time tank capacity. An audible buzzer will sound when full.



Flip the med level switch to on position



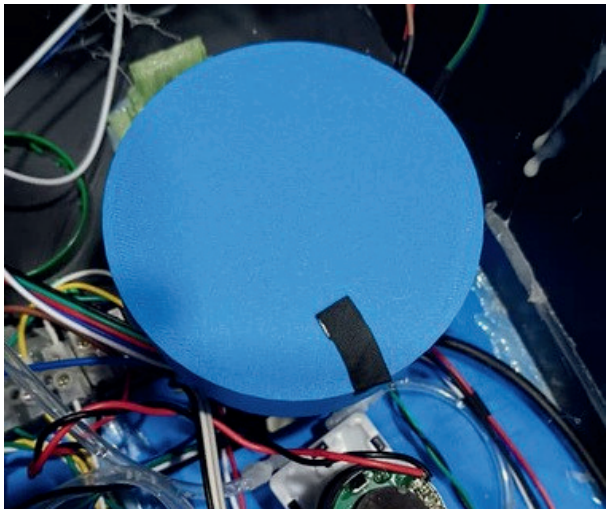
Ensure the Med Level progress bar indicates normal operational status.



This is the med solution tank fill in with med



Bar indicators reflect real-time liquid level during filling.

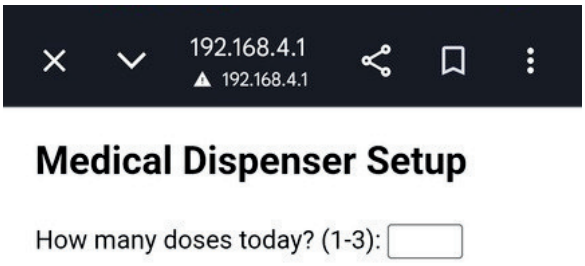




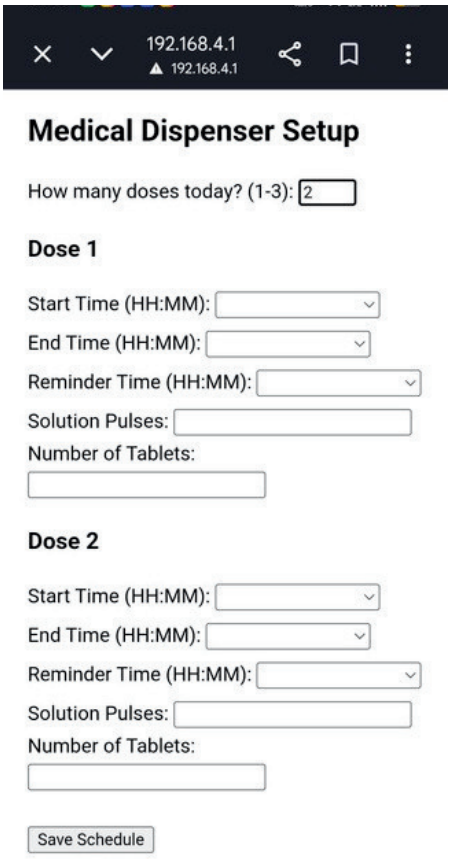
Open the lid and place the required pills/tablets directly into the rotating compartment of the pill dispenser module.



Open the Wi-Fi / WLAN settings on your smartphone or tablet.

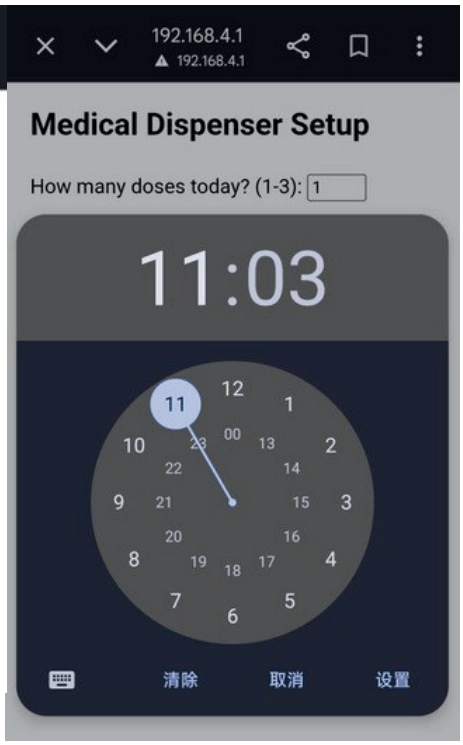


Access to the webpage and number of doses user take a day



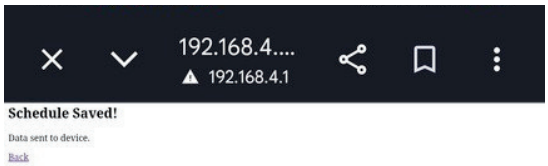


Selecting daily dose frequency expands 5 configuration parameters per dose (time window, reminder, pulses, pill count).



Click the boxes(time picker/time input box) to select the time range of taking each dose





After input all the data click save schedule